

Fundamentals of Research Methodology

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Outline

☐ What is needed to get a Ph.D. ?

- A systematic and planned approach to achieve something relevant and significant in a defined field of research in a comfortable means.

PART A

- ☐ What is a thesis?
- ☐ Choosing a Research Project
- ☐ Common Pitfalls and Hurdles
- ☐ Executing a research project.
- ☐ How to cope with Reading?

PART B

- ☐ Guidelines to Scientific Writing
 - LaTeX and BibTeX
 - Organizing your Research Thesis
 - ☐ Referencing / Referencing Standards
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What is Graduate Research ?

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- The distinguishing mark of graduate research is *an original contribution to knowledge*.
 - The thesis is a formal document whose sole purpose is to prove that you have made an original contribution to knowledge. Failure to prove that you have made such a contribution generally leads to failure.

 - To this end, your thesis must show two important things:
 - You have identified a worthwhile problem or question which has not been previously answered,
 - You have solved the problem or answered the question.
 - Your contribution to knowledge generally lies in your solution or answer
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What the Graduate Thesis is?

- ❑ Examiners read your thesis to find the answers to the following questions:
 - What is this student's research question?
 - Is it a good question? (has it been answered before? is it a useful question to work on?)
 - Did the student convince me that the question was adequately answered?
 - Has the student made an adequate contribution to knowledge?
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What is a Thesis?

A doctoral thesis must show *evidence of independent enquiry, originality* in the methods used and/or in the conclusions drawn and must make an *appreciable new contribution to knowledge* in the candidates field.

- How original and significant must Ph.D. research be?
 - The phrase 'make an appreciable new contribution to knowledge' suggests a simple rule of thumb. It should be possible to distil from the thesis, papers worthy of publication in a journal.
 - Supervisor with help of past theses in the area would help clarify this idea!
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Choosing a Research Project

❑ Most difficult task facing a new entrant!

Having chosen the general area or problem you want to work on, you must try to define a specific project. This is where your supervisor comes in.

Your supervisor may have some projects to suggest and will also be able to pass an opinion on the worthiness and do-ability of anything you suggest.

❑ Beginning students tend to underestimate the worthiness and overestimate the do-ability of projects

Quite modest sounding projects prove harder than they look.

Choosing a Research Project

Criteria for a Research Project

- You must be enthusiastic about it.
 - Solving the problems it entails must be worthy of a Ph.D.
 - It must be within sight of the state of the art, i.e. it must be 'doable' in three years.
 - There must be someone willing to supervise.
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Standard Pitfalls

- ☐ Solving the World
 - ☐ Manna from Heaven
 - ☐ Computer Bum
 - ☐ Yet Another Language
 - ☐ Micawberism
 - ☐ Ivory Tower
 - ☐ Misunderstood Genius
 - ☐ Lost in Abstraction
 - ☐ Ambition Paralysis
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Psychological Hurdles

- ☐ Mental Attitude
 - ☐ Research Impotence
 - ☐ Dealing with Criticism
 - ☐ Early Morning -- Cold Start
 - ☐ Theorem Envy
 - ☐ Fear of Exposure
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